Magpie — Sprint 19 spec: the evidence bank (comp memory)

Audience: Codex. **Author:** Claude. **Status: ready to implement. Locks:** D-128 (evidence bank), D-129 (evidence rule), D-130 (calculator), D-131 (sequencing — this is step 3). **Facts:** F-09 (no scraping), F-19 (2026 fee model), F-20 (no new eBay API usage — plan status still unconfirmed). **Builds on:** Sprint 14 (Comparable + capture + similar-matching), Sprint 18 (calculator v1 + `price_basis` + canonical normalisation, live at `f4deee3`).

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1. Goal

Make Magpie's **owned evidence instantly usable at the point of purchase**. Today the calculator's evidence lookup leans on existing items; at a market the item **doesn't exist in Magpie yet**. This sprint delivers **descriptor-based lookup** (category + a few key terms → matched own sales + approved comps, ranked and summarised), a **fast capture flow that grows the bank mid-session**, and a **"Bought it" step** that turns a good verdict into an inventory item with the right cost basis. The flywheel of D-128, made real: every capture makes the next market-stall answer faster.

2. Scope

IN:

- **Descriptor-based evidence lookup** (§4) — no item record required.
- **Evidence panel** on the buy calculator + item detail (§5).
- **Fast-capture v2**, phone-optimised, enriching the same lookup (§6).
- **"Bought it" → new Item** with cost basis (§7).
- **Descriptive strength/freshness labels** with stated rules (§8).
- **Folded polish from Sprint 18:** calculator **safe-area fix** (status-bar overlap) + the **formula-parity fixture** (§9).

OUT (explicitly):

- ❌ Any AI in the evidence path — no AI matching, ranking, or numbers (D-129; AI-assisted *search-term* help stays in the existing Sprint-15 feature, untouched).
- ❌ Any scraping/fetching or **new eBay API usage** (F-09; F-20 gate still in force).
- ❌ Stored lookup history — descriptor lookups are **transient** (nothing persisted unless the human captures a comp or creates an item).
- ❌ Profit-intelligence views (Sprint 20), lot manager (21), multi-channel (22).

3. Data model (additive, minimal)

- Reuse `Comparable` (+ `price_basis` from Sprint 18). Add only what matching needs if missing: a **category** tag and a small **keywords/attributes** field on `Comparable` so comps captured at a lookup are findable by descriptors later (additive migration; backfill nulls as-is).
- **No new lookup/session models.** "Bought it" creates a standard `Item` (existing model).

4. Descriptor-based evidence lookup

- An endpoint (+ calculator integration) taking **category + a few key terms/attributes** (e.g. "coin · 1966 · 50c · round") and returning matched evidence, **ranked: own** `SaleRecord` **exact → own similar → approved** `Comparable` **s** (extend the Sprint-14 matcher to descriptor input; deterministic, explainable matching — no fuzzy magic that can't be explained in a tooltip).
- **Summary stats per lookup:** low / median / high / count / newest date, computed **only from known-basis rows normalised to seller_receives** (D-129 + the Sprint-18 accepted behaviour). **Unknown-basis rows are returned and shown, visibly labelled "basis-uncertain," and excluded from the precise stats** — consistent with Sprint 18; do not change that behaviour.
- One-tap **"use this"** applies the median (or a tapped individual row's price) to the calculator's sell-price field, updating the **source label** accordingly (own-sale exact / own-sale similar / approved comp). What-if typing remains available and remains **never persisted** (regression from Sprint 18 stays green).

5. Evidence panel

- On the **buy calculator**: the lookup's matched rows + summary, each row showing **price (canonical basis), basis label, source tag, date, recency band**. Collapsed

by default on phone; one tap to expand.

- On **item detail**: the same panel driven by the item's own category/attributes (replacing/augmenting the current own-sales-first view without regressing it).

6. Fast-capture v2 (phone-first)

- From an open lookup: **"capture a comp"** prefills category/keywords from the lookup context; the human enters **price**, picks **basis** (quick-pick, **default** `unknown`), **source**, optional date/link/note. Target: **≤ ~4 taps + the price** on a phone.
- On save, the comp is a normal approved `Comparable` **and the current lookup refreshes immediately** — the user sees the bank grow mid-session.

7. "Bought it" → Item

- From a calculator result: one tap creates a **new** `Item` with `acquisition_cost` **= the price actually agreed** (prefilled from the asking field, editable), category + descriptors carried into the item's fields, and a note referencing the calc context (e.g. "created from buy calculator, MaxBuy $X, verdict BUY").
- The user lands on the item edit screen to finish (photos etc.). **Nothing beyond prefill is automatic**; no evidence rows are attached or created by this step.

8. Strength / freshness labels (honest, rule-stated)

- Per lookup: a descriptive line — **"n = <known-basis count> · newest <age>"** — plus a simple band, **STRONG / THIN**, with the rule **stated in a tooltip** (e.g. STRONG = ≥3 known-basis rows within 12 months including ≥1 within 90 days; otherwise THIN — implement exactly one clear rule and display it).
- **No invented confidence percentages or scores.** Counts, dates, and a transparent band only (no-false-precision, same spirit as D-129).

9. Folded polish (from Sprint 18 closure)

- **Safe-area fix:** the calculator's top field sat under the iOS status bar in the closure screenshot — respect the top safe-area inset (same treatment as the Sprint-16 pass) on `/buy-calculator`.
- **Formula-parity fixture (mandatory):** the fee/ROI formulas now exist in **both** backend and frontend (the Sprint-18 client-side what-if fix). Create **one shared**

canonical fixture — at minimum: the $76.92 free-selling case; one Pro-mode case (fees inside the division); one BPF-breakpoint round-trip — **asserted in BOTH the backend and frontend suites**. Rule (comment it in the fixture): any formula change updates the fixture and both suites must stay green.

10. Design

Sprint-16 bright high-contrast tokens; phone-first (this is the market-stall tool). Evidence rows legible at 390 px; big tap targets; the lookup → use-this → verdict loop should take seconds. Reduced motion; no overflow.

11. Acceptance criteria (closure gate)

- **Lookup:** descriptor lookup ranks own-sale exact → similar → approved comps (tested); stats computed only from known-basis rows on the canonical basis (tested); unknown rows shown + labelled + excluded (tested); "use this" applies price + correct source label.
- **Capture v2:** creates a properly tagged `Comparable` (basis default `unknown`) and the **same lookup refreshes to include it** (test the round-trip).
- **Bought it:** creates an `Item` with the correct `acquisition_cost` + carried descriptors (tested); nothing else auto-created.
- **Transience:** lookups persist nothing by themselves; what-ifs remain never-persisted (Sprint 18 regression stays green).
- **Parity fixture:** the shared canonical cases pass in **both** suites.
- **Safe-area:** no content under the status bar on `/buy-calculator` (phone screenshot).
- **No new network calls** (assert; F-09/F-20).
- **Process:** backend + frontend tests green; **dual-lane Validation green**; additive migrations only; **pre-migration encrypted backup + restore spot-check** (new Comparable fields); rebuild + restart (Administrator, per the known service permissions); **phone screenshots are headline evidence** (lookup → use-this → verdict; capture round-trip). Evidence as `SPRINT19_EVIDENCE.md`.

12. For Codex

Reasoning: High. The matcher must be **deterministic and explainable** — no opaque

scoring. **Hard lines:** evidence math consumes **only in-Magpie rows** (D-129 — keep the existing guard honest for the new path); **no AI anywhere in this sprint's paths**; **no scraping and no new eBay API calls** (F-09/F-20); lookups are **transient**; what-ifs never persist. Keep Sprint 18's accepted unknown-basis behaviour exactly (visible, labelled, excluded from precise stats). Additive migrations; pre-migration backup + restore spot-check. **Never print/commit secrets** — the credential rotation is still in progress, so never echo current `.env` values in output or diagnostics. Evidence as `SPRINT19_EVIDENCE.md`. Next per D-131: Sprint 20 = profit intelligence v1.